Problem-Solution Fit

Date	28 June 2026
Team ID	xxxxxx
Project Name	Raising waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • **Disaster Management Authorities** • **Meteorologists** • **Government Agencies** • **Flood-Prone Communities**	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** • **Limited access to real-time weather data.** • **Manual monitoring takes time.** • **Budget and technology constraints.** • **Lack of advanced prediction tools.**	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** **Current Solutions: Traditional forecasting and manual monitoring.** **Pros:** • **Easy to implement.** • **Widely used.** **Cons:** • **Lower prediction accuracy.** • **Delayed warnings.** • **Limited real-time analysis.**

2. PROBLEMS / PAINS + ITS FREQUENCY **[PR]**

- **Delayed flood warnings during monsoon.**
- **Inaccurate flood prediction methods.**
- **Loss of lives and property due to late response.**
- **Occurs frequently in flood-prone regions.**

9. PROBLEM ROOT / CAUSE [RC]

☐ **Dependence on traditional forecasting.**
☐ **Insufficient analysis of historical weather data.**
☐ **Lack of accurate machine learning prediction.**
☐ **Delayed decision-making.**

7. BEHAVIOR + ITS INTENSITY **[BE]**

☐ Regularly monitors weather updates.
☐ Responds quickly during heavy rainfall.
☐ Frequently checks flood alerts in monsoon.

3. TRIGGERS TO ACT [TR]

- **Heavy rainfall.**
- **Rising river water levels.**
- **Extreme weather forecasts.**
- **Flood alerts from weather departments.**

4. EMOTIONS BEFORE / AFTER **[EM]**

Before: Worried, uncertain, unprepared.
After: Confident, informed, prepared.

10. YOUR SOLUTION [SL]

☐ **Develop a Machine Learning Powered Flood Prediction System using Decision Tree, Random Forest, KNN, and XGBoost. The best-performing model is integrated into a Flask web application to provide accurate flood risk predictions, enabling authorities to issue early warnings and improve disaster preparedness.**

8. CHANNELS OF BEHAVIOR [CH]

ONLINE
☐ Weather websites
☐ Flask web application
☐ Government portals
☐ Mobile notifications

OFFLINE
☐ Disaster management offices
☐ Emergency control rooms
☐ Local administration
☐ Public announcements